

Course: Chemistry

Course Code: Chem-2301

Segment-2

"Electronic Theory of Elements"

Segment Outline:

- Chemical bonds
- Types of Chemical bonds
- Ionic, Covalent & Coordination bonds
- Hybridisation of atomic orbitals
- Bonding in simple molecules
- Elementary idea about MOT

Def Chemical Bond:

A Chemical bond is the association of atoms or ions to form molecules, crystals and other structures.

The bond may result from the electrostatic force between oppositely charged ions as in ionic bonds or through the sharing of electrons

as in covalent bonds or some combination of these effects.

⇒ Principles of Electronic Theory of Chemical Bond:

1. Chemical combination between atoms of the same or different elements take place due to the tendency by the outermost electron groups to attain the stable electronic Confⁿ of inert gases.
2. The attainment of inert gas electronic configuration may take place by complete transference of electron from one atom to another. The resulting electrically
3. The attainment of inert gas electronic Confⁿ may occurred by sharing of electrons (in pairs) betⁿ two atoms. The Chemical bond so formed is known as covalent bond.

4. The attainment of electronic confⁿ of inert gases may also happen by sharing of both the electrons from one atom. The bond so formed is known as Coordination Covalent bond.

~~4~~ Some Relevant Terms about Chemical Bonds:

a. Atomic Number: The number of protons exist in the nucleus of an atom is called atomic number of that element.

e.g., Atomic number of Na is 11.

Atomic number of Ca is 20.

In general,

$$\begin{aligned}\text{Atomic Number} &= \text{Number of Protons} \\ &= \text{Number of electrons.}\end{aligned}$$

b. Electronic Configuration: The arrangement of electrons in orbitals around an atomic nucleus is called electronic confⁿ of element.

: **Orbitals** are filled up with the electrons according to the Aufbau Principle.

→ The Orbitals are filled up with electrons in the increasing order of their energy.

→ The Orbitals of minimum energy are filled up first with electrons and then the Orbitals of higher energy start to fill.

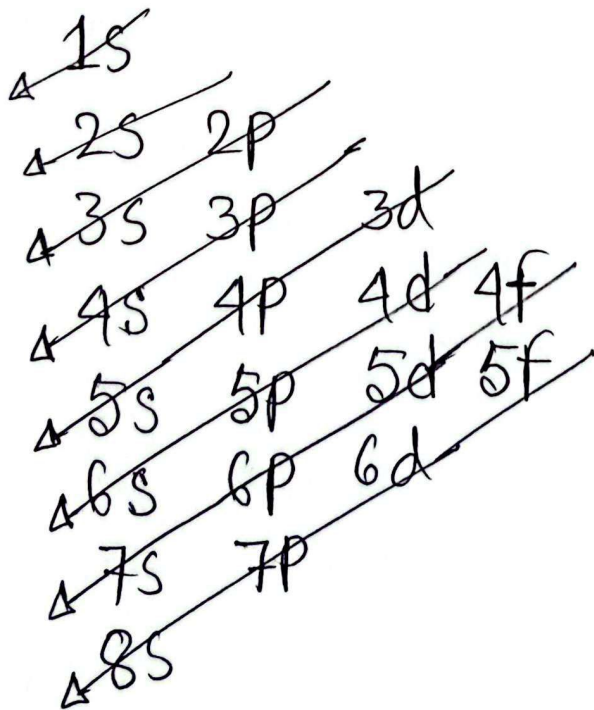
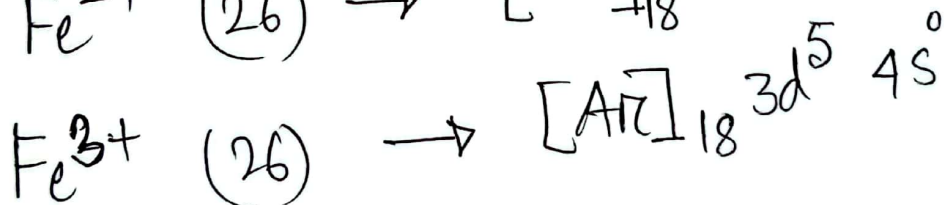
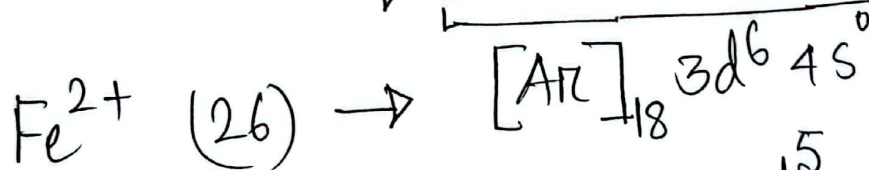
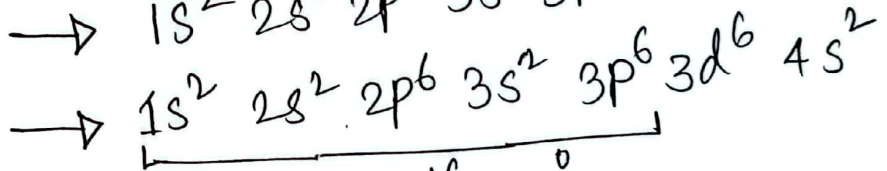
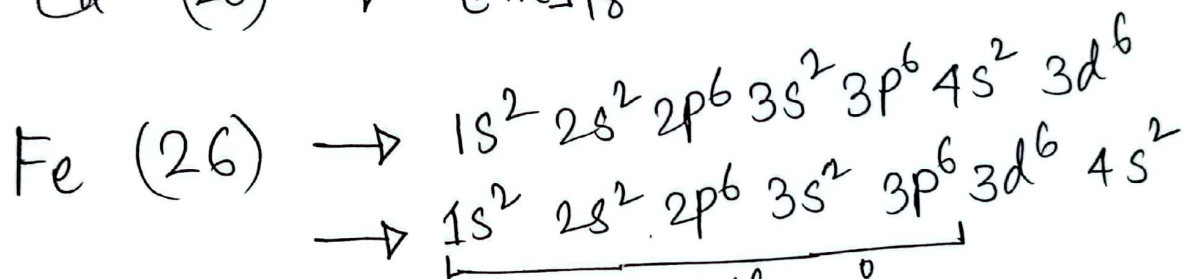
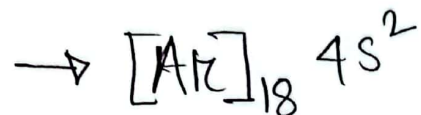
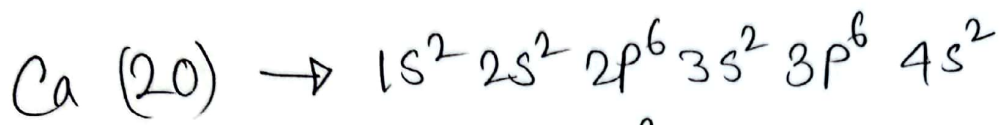
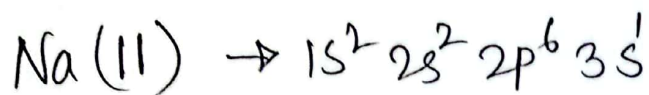


Fig: Sequence of Orbitals energy
(Aufbau Principle)

Let us show electronic confn of some elements and ions.

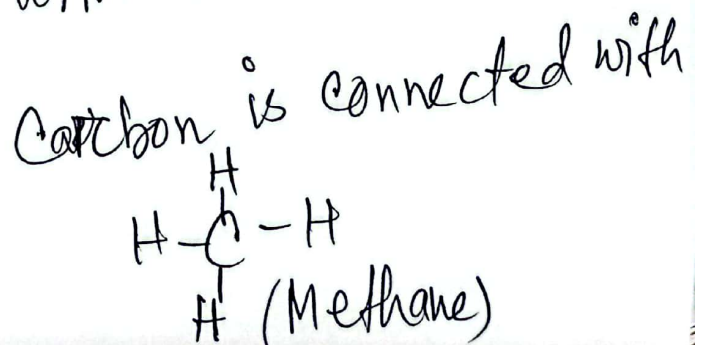


$\Rightarrow$ Valency: Valency is the combining ability of elements to form compounds.

e.g., Valency of Carbon (C) is 4.

Carbon can connect with 4 monovalent atoms.

In CH_4 (Methane)
4 Hydrogen atoms.

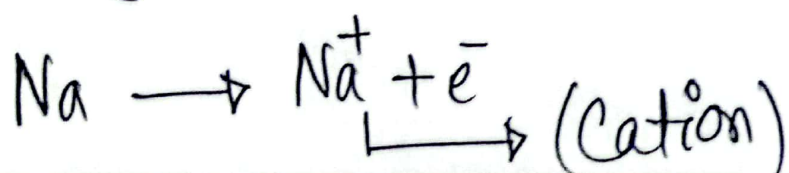


⇒ Metals and Non-metals: Elements which outermost energy level has 1, 2 or 3 electrons, generally they may be metals (except some exceptional) and elements which outermost energy level has 4, 5, 6 or 7 electrons generally they may be non-metals.

We should know -

Metal + Non-metal = Ionic Bond
Non-metal + Non-metal = Covalent Bond
Metal + Metal = Metallic Bond.

⇒ Ion formation: When atoms get or lose its outermost shell's electrons they turn into positive charged ions (cations) or negative charged ions (anions)



⇒ Different Types of Chemical Bond:

→ There are 4 types Strong Chemical bonds.
They are -

1. Ionic or Electrovalent Bond
2. Covalent Bond
3. Co-ordinate Covalent Bond
4. Metallic Bond.

→ There are 2 types Weak Chemical bonds.
They are -

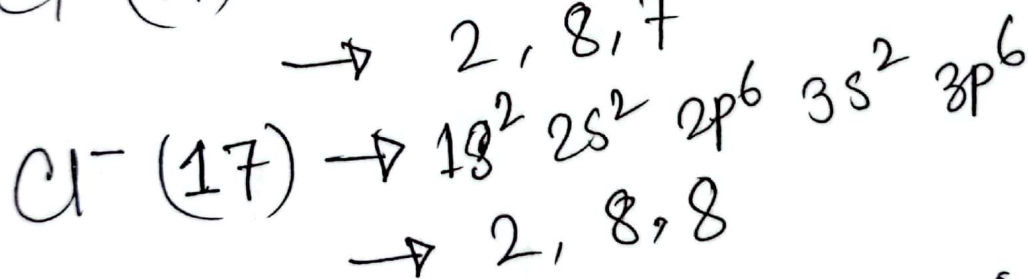
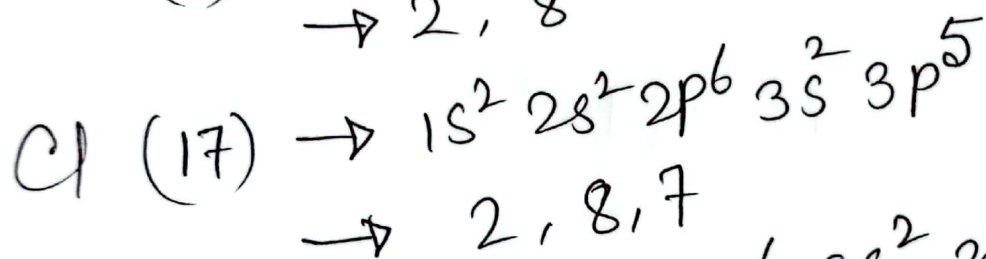
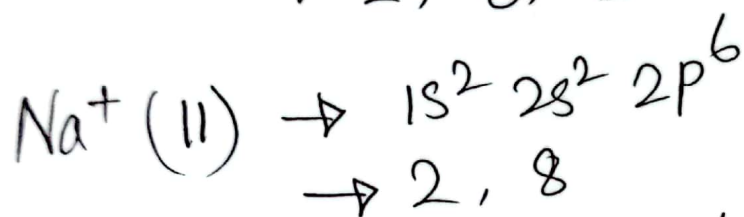
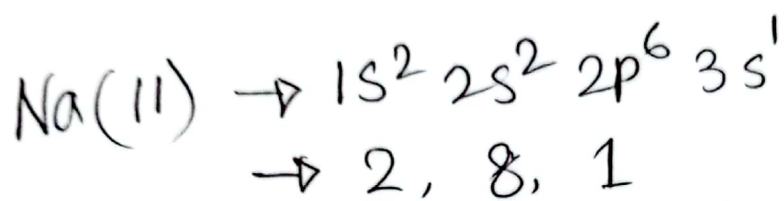
1. Hydrogen Bond
2. Van der Waals Interaction

Ionic or Electrovalent Bond

(Bonding by transference of electrons)

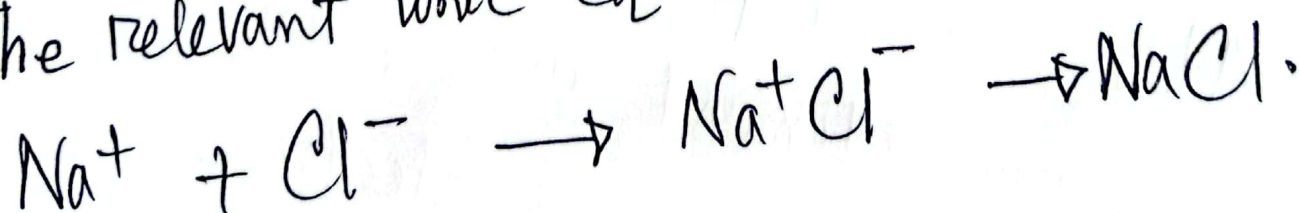
The Chemical bond which is formed by the transference of electron from one atom to other is called ionic bond or electrovalent bond.

Thus Na atom acquires the configuration of Ne (2,8) and forms Na^+ and Cl atom acquires the configuration of Ar (2,8,8) and forms Cl^- ion.



The resulting ions (cation and anion) are held together by electrostatic force of attraction and form NaCl.

The relevant ionic eqⁿ is shown below.



In ionic compound, one of the combining atoms has excess of electrons than the stable number (2 or 8) in its valence-shell. While the other atom has short of electrons and hence needs electrons to complete its Octet or duet.

They form cations and anions and these ions combine by the electrostatic force of attraction. Thus resulting compound is known as ionic compound and bond exist in the compound is termed as ionic compound.

Example of an ionic compound formation is discussed below.

Let us discuss about the formation NaCl . In this compound, Na atom transfers its excess one electron to Cl atom.

⇒ General Properties of the Ionic Bond:

1. Form crystalline compound.
2. Conduct electricity
3. Have high melting and boiling points
4. High Chemical reactivity
5. More soluble in polar than non-polar solvents.

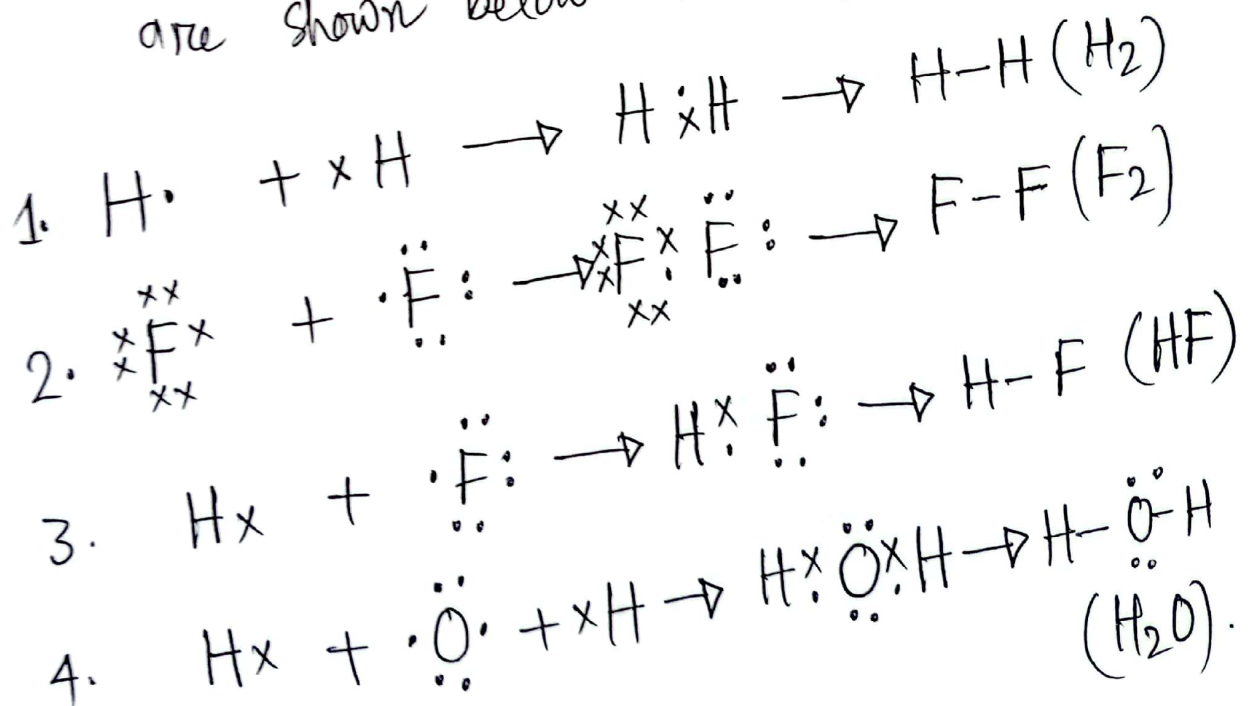
Covalent Bond

Lewis Theory: The covalent bond is formed by the sharing of electrons by both the atoms equally. Thus formed compound is called covalent compound.

Atoms of different elements or atoms of same element may form covalent bond by sharing of electrons equally.

Sharing of electrons between atoms always occurs in pairs and in each pair of shared electrons one is contributed by each atom. One pair of electrons corresponds to a single bond, two pairs to double bond and three pairs of shared electrons to triple bond.

Examples of some covalent molecules & compounds are shown below according to Lewis concept.

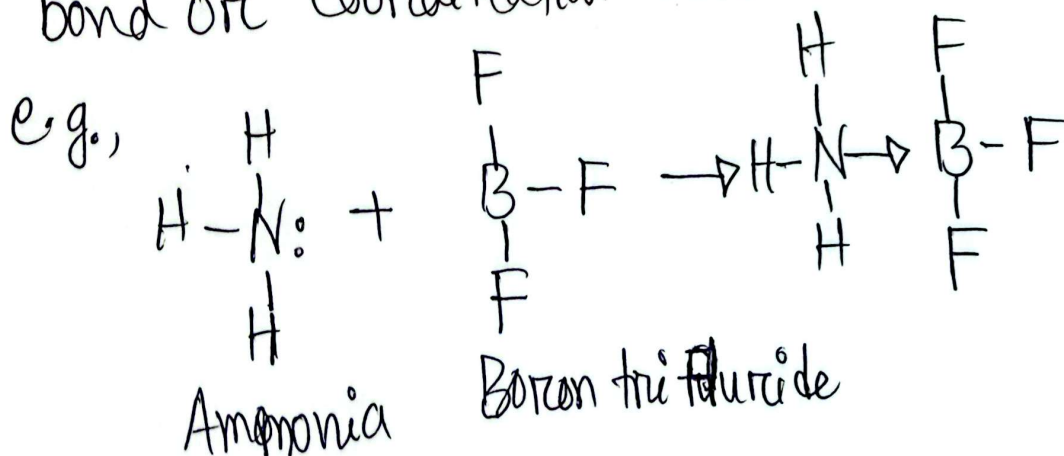


⇒ General Properties of Covalent Compounds:

1. Covalent compounds are generally exist in gaseous or liquid state.
2. May form Crystalline compounds.
3. Generally covalent compounds are soluble in non-polar solvents.
4. Slow Chemical reactivity
5. Some covalent compounds conduct electricity.

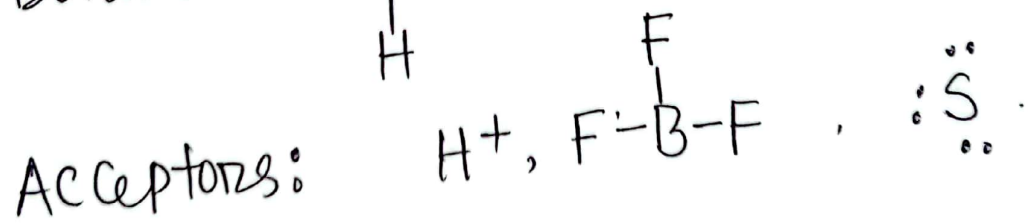
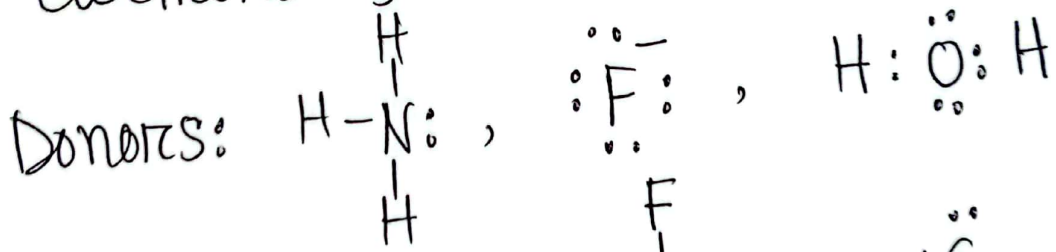
Coordination Bond

The covalent bond which form by the sharing of both electrons (electron pair) from one atom to another atom is called coordination bond or coordination covalent bond.



⇒ Necessary Conditions for the formation of coordination bond:

1. One atom, molecule or ion must have an pair of electron available for sharing with other atom by donation.
2. One atom, molecule or ion must have capability of sharing the lone pair of electrons by acceptance.



Hybridisation

Hybridisation is a process in which two or more atomic orbitals of valence shell of an atom mix up to form the same number of identical orbitals of equal energy.

⇒ Different Types of Hybridisations:

There may have different types of hybridisation of orbitals. Such as:

1. sp^3 hybridisation.

3 structures
Tetrahedral

2. sp^2 "

Plane trigonal

3. sp "

Linear

4. sp^2d "

Square Planar

5. sp^3d "

Trigonal bipyramidal

6. sp^3d^2 "

Octahedral

7. sp^3d^3

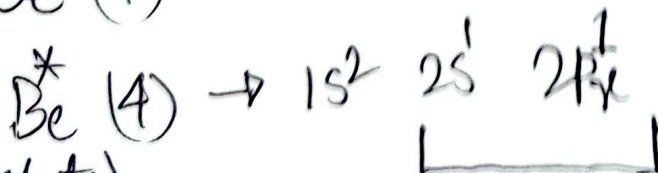
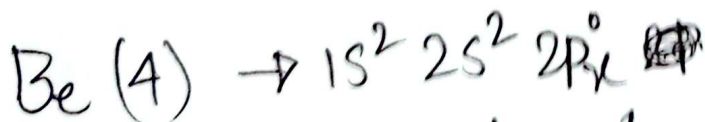
Pentagonal bipyramidal

8. d^2sp^3 or sp^3d^2

Octahedral.

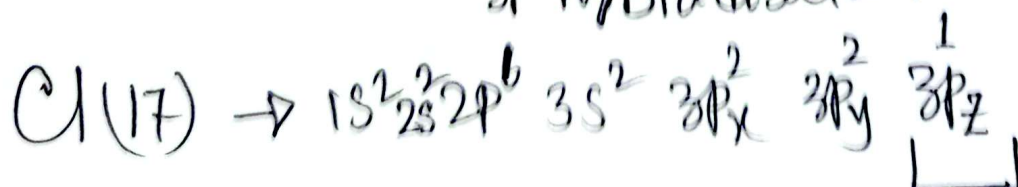
Some Examples

1. Formation $BeCl_2$ compound by sp hybridisation



(Excited state)

sp hybridisation



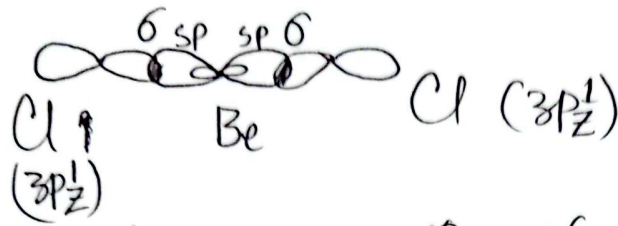
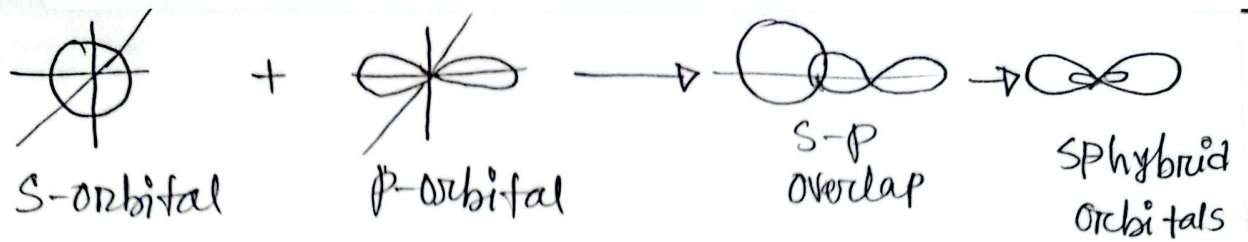
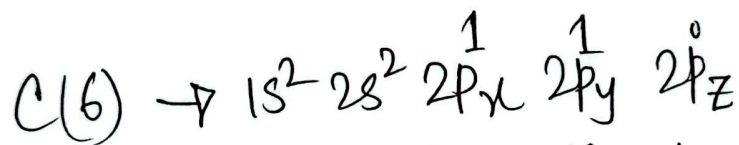
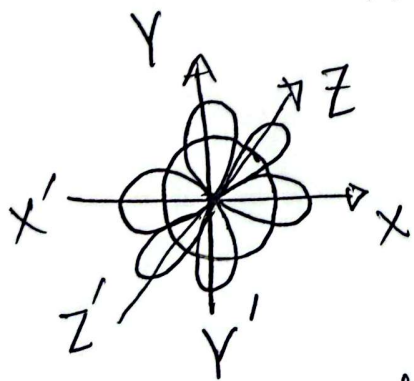
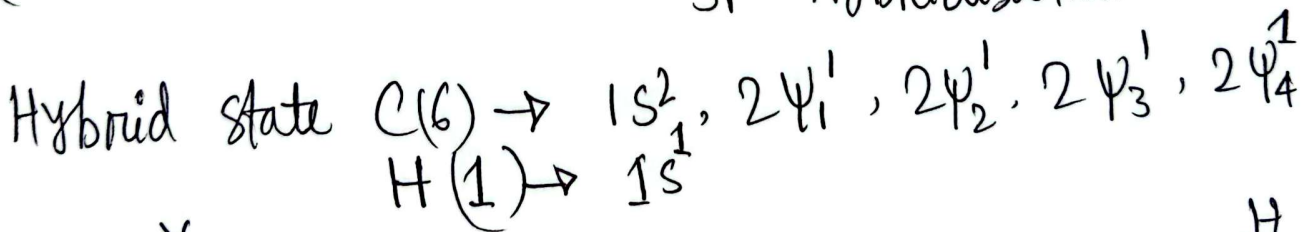
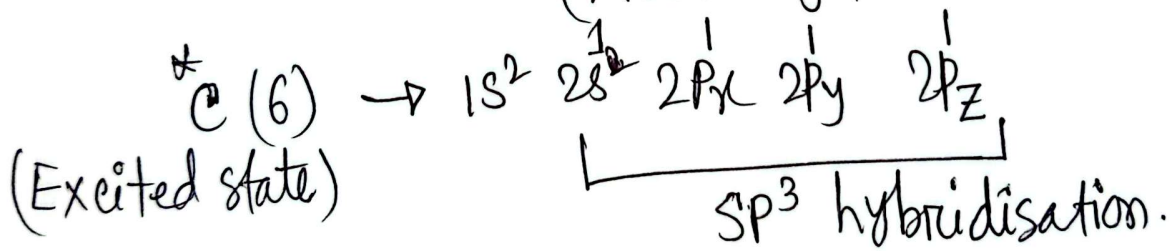


Fig: Formation of BeCl₂ by sp hybridisation

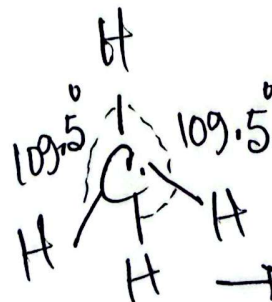
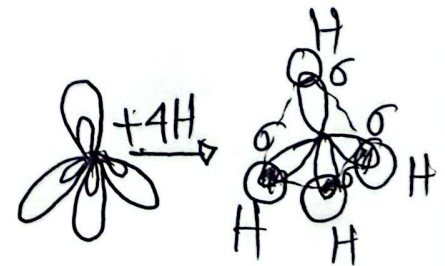
2. Formation of CH₄ by sp³ hybridisation.



(According to Hund's rule)



sp³ hybridisation



Struct. of CH₄

$\rightarrow$ Tetrahedral.